Convolutional VAEs for Chest X-ray Generation

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Abstract

We train convolutional VAEs on pediatric chest X-rays and score samples with FID and Inception Score. A sweep over the KL weight $\beta \in \{1, 0.1, 0.01\}$ shows a trade-off: small β reconstructs better (FID 206 vs. 252) but samples worse from the $\mathcal{N}(0, I)$ prior (323 vs. 300). Sampling from a Gaussian mixture fitted to the training latents removes most of the gap; the best model reaches FID 214.

1 Introduction

We implement a VAE Kingma and Welling [2013] in PyTorch and train it on the Chest X-Ray Images (Pneumonia) dataset Kermany et al. [2018]. The closest prior work we found Chandra et al. [2025] trains a similar 64×64 VAE on this data with a very small KL weight (β=0.005) and judges samples by eye. We add FID Heusel et al. [2017] and Inception Score Salimans et al. [2016], and measure how β and the sampling distribution change them.

2 Method

Model. Encoder: four stride-2 4×4 convolutions (64, 128, 256, 512 channels) with BatchNorm and LeakyReLU(0.2), then two linear heads for the mean and log-variance (clamped to $[-10, 10]$) of a 128-d Gaussian posterior. The decoder mirrors it with transposed convolutions and a sigmoid output (8.7M parameters).

Loss and sampling. $\mathrm{BCE}(x, \hat{x}) + \beta\, D_{\mathrm{KL}}(q(z|x) \,\|\, \mathcal{N}(0, I))$, BCE summed over pixels; β grows linearly from 0 over the first 10 epochs to avoid posterior collapse. "Prior" samples decode $z \sim \mathcal{N}(0, I)$. Since small β leaves the aggregate posterior far from $\mathcal{N}(0, I)$, "GMM" samples instead decode z from a 10-component Gaussian mixture fitted to the training latent means Ghosh et al. [2019].

3 Setup

Data. The 5,216 training images are split 90/10 (seeded) into 4,694 train and 522 validation images; the 624 test images are the FID reference. Images are grayscale, resized and center-cropped to 64×64, scaled to $[0, 1]$; training augmentation is rotation ($\pm5^\circ$) and translation (5%), without flips (they would mirror the anatomy).

Training and evaluation. Adam, lr 10^{-3} (halved on validation plateaus), batch 128, 100 epochs, seed 0, on an Apple M1 Pro GPU (≈8 s/epoch); the lowest-validation-loss checkpoint is evaluated. FID and IS use 2,000 generated images, PyTorch-Ignite[1], and ImageNet InceptionV3 Szegedy et al. [2016] features (grayscale replicated to 3 channels, resized to 299×299); IS is the mean±std over 10 splits. "Recon." is the FID of the 624 test-set reconstructions, a reference point. FID is biased upward for small sample sizes, so "GMM@624" scores only 624 GMM samples, to compare like with like against Recon.

[1] https://github.com/pytorch/ignite

Preprint.

Table 1: FID (lower is better) against the 624 test images. Prior and GMM use 2,000 generated images; GMM@624 and Recon. use 624. Inception Score (higher is better) is for prior samples.

β	FID prior	FID GMM	FID GMM@624	FID recon.	IS prior
1	299.7	248.5	256.8	252.2	1.46 ± 0.04
0.1	312.2	231.9	233.7	226.6	1.47 ± 0.03
0.01	322.7	214.2	217.9	205.6	1.36 ± 0.02

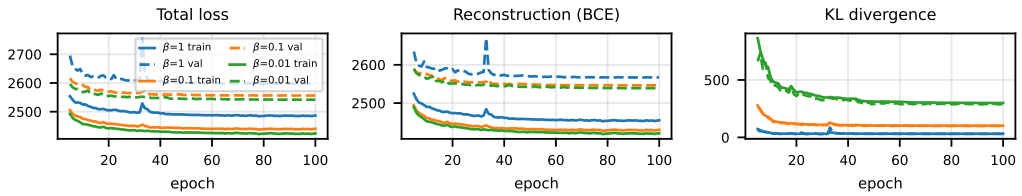


Figure 1: Training (solid) and validation (dashed) losses; epochs 1–4 are omitted because the KL warm-up makes them orders of magnitude larger.

4 Results

Training. All runs converge smoothly (Fig. 1). Smaller β lowers validation BCE (2567, 2546, 2539 for $\beta = 1, 0.1, 0.01$) but raises KL (31, 99, 289 nats): codes carry more information but drift from the prior.

Quantitative. Table 1 shows the trade-off. Reconstruction FID improves as β falls (252, 227, 206) while prior-sample FID worsens (300, 312, 323). GMM sampling reverses this (248, 232, 214 with 2,000 samples). With matched size, Recon. $<$ GMM@624 $<$ prior holds for every β (GMM@624: 257, 234, 218), so GMM sampling closes most, not all, of the gap between prior samples and reconstructions: the gap is mostly prior mismatch, not decoder quality. IS of prior samples is 1.36–1.47; ImageNet classes separate these images poorly, so we treat IS as uninformative. Absolute FIDs are high (blurry images, 624 real images) and not comparable to other datasets.

Qualitative. Prior samples of the β=0.01 model (Fig. 2, left) are noisy textures with dark borders (likely the rotation augmentation's corner fill), while GMM samples (middle) show plausible rib cages and lung fields, matching the FID gap. Reconstructions keep global shape but lose rib detail. A PCA of the test latents separates NORMAL and PNEUMONIA only partly (heavy overlap), so we do not claim a clean disease separation. Limitations: blurry output, one seed per setting, untuned GMM size, no clinical or downstream validation.

AI usage statement

Claude Code (Anthropic; Claude Opus 5.5 and Claude Sonnet 5.5) wrote the code and tests, ran training and evaluation on the author's laptop, searched for and summarized related work and checked the references, and drafted this report. A separate Claude Fable 5.1 review of the code and report led to two fixes (interrupt handling; the FID sample-size comparison). The author chose the setup, approved the design and plan, and approved each download.

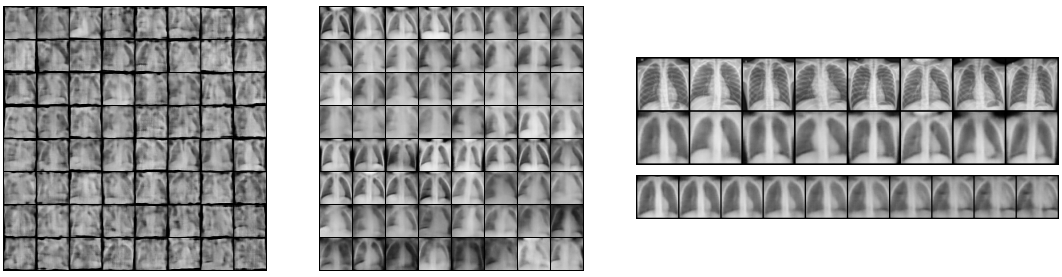


Figure 2: β=0.01 model. Left: prior samples. Middle: GMM samples. Right: test images over their reconstructions (top); latent interpolation from a normal to a pneumonia image (bottom).

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